

PROJECT HELIX

Helix

RISC-V CPU | 5-Stage Simulation Report

Vivado 2025.1 xsim behavioral regression | 5 stages | CPI = 5

DOCUMENT	Helix 5-CPI Simulation Report
PROJECT	Helix (RISC-V CPU, 5-CPI)
IMPLEMENTATION	5-Stage Pipeline
SIMULATOR	Vivado 2025.1 xsim
REPORT DATE	June 2026
SOURCE PDF	ARIA-32_5Stage_SimReport.pdf

1

Simulation Summary

13 / 13 testbenches

PASS

STATUS

5-Stage Pipeline

IMPLEMENTATION

5

CPI

Vivado 2025.1 xsim

SIMULATOR

This report summarizes the behavioral simulation regression for the Helix 5-stage sequential pipeline. The workload and checks match the single-cycle suite, with halt detection adapted for WB-stage architectural PC updates.

5-Stage Sequential Pipeline Flow

STAGE 0

IF

Fetch IMEM[pc], tentatively advance PC, fill IF/ID.

STAGE 1

ID

Decode instruction, read registers, fill ID/EX.

STAGE 2

EX

ALU, compare, branch target, fill EX/MEM.

STAGE 3

MEM

LDR/STR data memory access, fill MEM/WB.

STAGE 4

WB

Commit register result and architectural PC.

One instruction is advanced through IF, ID, EX, MEM, and WB; architectural PC and register writes commit in WB.

Regression Status Matrix

Testbench	Scope	Timeout	Cycles	Status
tb_cpu	ISA integration	2500 cycles	425	PASS
tb_memcpy	10-word memory copy loop	2500 cycles	330	PASS
tb_array_sum	$C[i] = A[i] + B[i]$, $N=25$	4000 cycles	1150	PASS
tb_minmax	Min/max linear scan, $N=25$	3500 cycles	950	PASS
tb_sort	Bubble sort ascending, $N=25$	15000 cycles	9000	PASS
tb_factorial	$0!$ through $12!$ via BL/BX	7000 cycles	3750	PASS
tb_bitops	AND/OR/XOR/NOT/LSL/LSR x 8 inputs	2500 cycles	800	PASS
tb_gcd	Euclidean GCD x 6 pairs	5000 cycles	3000	PASS
tb_power	base^{exp} binary exponentiation x 8	6000 cycles	2250	PASS
tb_isqrt	$\text{floor}(\text{sqrt}(n)) \times 10$	18000 cycles	7500	PASS
tb_collatz	Collatz steps-to-1 x 8	20000 cycles	9000	PASS
tb_fibonacci	$F(n)$ iterative x 8	5000 cycles	2400	PASS
tb_bsearch	Binary search 16 queries / 150 elements	30000 cycles	14000	PASS

2 RTL Source Files

The 5-stage implementation preserves the single-cycle ISA modules while adding the stage FSM and IF/ID, ID/EX, EX/MEM, and MEM/WB pipeline registers inside `cpu_top.v`.

RTL File	Role in Simulation
<code>cpu_top.v</code>	Top-level: 5-stage FSM, pipeline registers IF/ID to ID/EX to EX/MEM to MEM/WB, APB slave, interrupt controller
<code>control.v</code>	Instruction decode to control signals (identical to single-cycle)
<code>alu.v</code>	32-bit ALU: ADD/SUB/MUL/AND/OR/XOR/NOT/LSL/LSR/CMP (identical)
<code>reg_file.v</code>	16x32 register file (flat signals <code>r0..r15</code> ; write gated to WB stage)
<code>cond_check.v</code>	Condition code evaluator (identical to single-cycle)
<code>inst_mem.v</code>	256x32 instruction SRAM, APB write-loaded at reset (identical)
<code>data_mem.v</code>	256x32 data SRAM (write enabled only at MEM stage, <code>stage==3</code>)
<code>defines.v</code>	ISA constants: format codes, opcodes, condition codes (<code>`include</code>)

Pipeline Stages

Stage	Index	Operation	Pipeline Register
IF	0	Read IMEM[pc] and advance PC tentatively	ifid_pc, ifid_inst
ID	1	Decode ifid_inst and read register file	idex_* (control, operands, rd)
EX	2	Run ALU operation and compute branch target	exmem_* (alu_result, branch)
MEM	3	DMEM read for LDR or write for STR	memwb_* (data, alu_result)
WB	4	Write result to register file and update architectural PC	pc <= arch, r0..r15 if rd != 0

Simulation Method

Area	Method
Program loading	APB writes to IMEM with rst_n=0 and prst_n=1; then rst_n=1 starts CPU execution
Halt detection	Sample dut.pc only when dut.stage == 3'd4; halt when pc_curr == pc_prev
Why stage==4 only	dut.pc is architectural only at WB; other stages hold stale or speculative values
Simulation end	\$stop after verify() at halt; xsim stays alive for close_simulation
Timeout	\$stop after each TB-specific watchdog; all timeouts are scaled about x5 relative to single-cycle
VCD dump	\$dumpfile / \$dumpvars at module start
Result check	Display PASS/FAIL per expected DMEM word or register value
IRQ in tb_cpu	irq pulsed at cycle ~500 in SYS mode; handler is single RETI
\$stop vs \$finish	All testbenches use \$stop so Vivado close_simulation can release simulate.log before the next TB launches in run_all.tcl

Architectural PC updates only at stage==4 (WB). Testbenches therefore gate halt detection with if (dut.stage == 3'd4); sampling dut.pc in other stages can read stale or speculative values.

Pipeline Properties

Property	Behavior
Hazard handling	None: one instruction in flight at a time through the sequential FSM
Branch resolution	Resolved at EX stage (stage==2); PC corrected at WB (stage==4)
DMEM write	Gated by stage==3 && mem_write; no false writes at other stages
Register file	Write gated to stage==4 && rd!=0; reads always available
Pipeline registers	IF/ID, ID/EX, EX/MEM, MEM/WB; each contributes one cycle of latency
Timeout scaling	All timeouts scaled about x5 relative to the single-cycle suite for CPI=5

4 Detailed Testbench Results

Each testbench is the single-cycle regression ported to the sequential pipeline. Program load, result checks, and expected values are unchanged; halt sampling is WB-gated.

01 tb_cpu

PASS

Source	tb/tb_cpu.sv	Timeout	2500 cycles
Cycles	425	Halt	PC sampled only when dut.stage == 3'd4 (WB).

General ISA integration test. Exercises all major instruction categories in one straight-line program: arithmetic, bitwise, branch taken/not-taken, load/store round-trip, INC/DEC, PUSH/POP, BL/BX subroutine call-return, and IRQ vector table (RETI). Ported from single-cycle; halt detection gates PC sampling on stage==4 (WB).

INSTRUCTION COVERAGE

MOVI ADD R-type SUB R-type MUL R-type AND OR XOR NOT LSL LSR CMP BGT BNE STR LDR INC
DEC PUSH POP BL BX B AL RETI

PASS CRITERIA

- R2 = 15 (ADD R0+R1=13, then +2 INC from sub_add2 subroutine via BL)
- R5 = 3055 (0xBEF from MOVI after BGT taken; FAIL path would set R5=0)
- R6 = 3055 (LDR from DMEM[R0] after STR; round-trip verify)
- R15 = 3055 (POP'd from stack; must equal R5)
- SP = SP_INIT (stack balanced after PUSH/POP)

NOTES

- Timeout x5 vs single-cycle (CPI=5). IRQ pulsed at cycle ~500 in SYS mode. IRQ handler: single RETI.

02 tb_memcpy**PASS**

Source	tb/tb_memcpy.sv	Timeout	2500 cycles
Cycles	330	Halt	PC stable at WB stage (stage==4) after counter reaches 0.

Memory copy loop copies 10 words from DMEM[0x100..0x124] to DMEM[0x200..0x224].

INSTRUCTION COVERAGE

MOVI LDR STR ADD immediate offset DEC BNE

PASS CRITERIA

- DMEM[0x200..0x224] matches source for all 10 words
- R0 = 0x128, R1 = 0x228, R2 = 0

NOTES

- Cycle count: 330 in 5-stage versus about 66 single-cycle. All 10 words PASS.

03 tb_array_sum**PASS**

Source	tb/tb_array_sum.sv	Timeout	4000 cycles
Cycles	1150	Halt	PC stable at WB stage after N=25 elements complete.

Vector addition computes $C[i] = A[i] + B[i]$ for N=25 elements.

INSTRUCTION COVERAGE

MOVI LDR ADD R-type STR INC CMP BNE

PASS CRITERIA

- DMEM[192..288] contains all $C[i] = A[i] + B[i]$ results
- Example: $A[3]=951 + B[3]=629 = 1580$ at DMEM[204]
- Example: $A[22]=743 + B[22]=854 = 1597$ at DMEM[280]

NOTES

- Cycle count: 1150 in 5-stage versus about 230 single-cycle. All 25 elements PASS.

04 tb_minmax**PASS**

Source	tb/tb_minmax.sv	Timeout	3500 cycles
Cycles	950	Halt	PC stable at WB stage after N=25 elements complete.

Linear scan finds the minimum and maximum values across a 25-element array.

INSTRUCTION COVERAGE

MOVI LDR CMP BLT BGT STR INC BNE

PASS CRITERIA

- DMEM[200] = 18
- DMEM[201] = 951

NOTES

- Array values include {427,83,612,951,38,776,245,18,534,889,...,667}; single-cycle PASS.

05 tb_sort

PASS

Source	tb/tb_sort.sv	Timeout	15000 cycles
Cycles	9000	Halt	PC stable at WB stage after sort loops complete.

Bubble sort orders 25 words in ascending order.

INSTRUCTION COVERAGE

MOVI LDR CMP BLE STR INC DEC BNE ADD

PASS CRITERIA

- DMEM[64..160] sorted ascending
- Expected order begins [18, 29, 38, 57, 83] and ends with 951

NOTES

- Complexity: $N*(N-1)/2 = 300$ worst-case compare-swap iterations x CPI=5.
- Worst case is about 300×8 instructions x CPI5 = 12000; timeout is 15000.

06 tb_factorial

PASS

Source	tb/tb_factorial.sv	Timeout	7000 cycles
Cycles	3750	Halt	PC stable at WB stage after n=0..12 sequence completes.

Iterative factorial test computes n=0..12 through a BL/BX subroutine flow.

INSTRUCTION COVERAGE

MOVI MUL STR BL BX R14 PUSH POP TST BEQ DEC BNE

PASS CRITERIA

- DMEM[128] = 1 (0!)
- DMEM[129] = 1, DMEM[130] = 2, DMEM[131] = 6, DMEM[132] = 24
- DMEM[133] = 120, DMEM[134] = 720, DMEM[140] = 479001600 (12!)

NOTES

- BL/BX uses LR for subroutine return.

07 tb_bitops

PASS

Source	tb/tb_bitops.sv	Timeout	2500 cycles
Cycles	800	Halt	PC stable at WB stage after all eight input groups complete.

Bitwise regression runs AND, OR, XOR, NOT, LSL, and LSR over eight input values, producing 48 outputs.

INSTRUCTION COVERAGE

MOVI LDR AND OR XOR NOT LSLI LSRI STR INC CMP BNE

PASS CRITERIA

- For each input: AND val&0x0F, OR val|0xF0, XOR val^0xFF, NOT ~val
- For each input: LSL val<<4 and LSR val>>4
- 48 result words are written at DMEM[128..303]

08 tb_gcd**PASS**

Source	tb/tb_gcd.sv	Timeout	5000 cycles
Cycles	3000	Halt	PC stable at WB stage after all six GCD pairs complete.

Euclidean subtraction GCD is evaluated across six input pairs using BL/BX subroutine return.

INSTRUCTION COVERAGE

MOVI LDR SUB R-type CMP BEQ BGT BL BX R14 PUSH POP STR

PASS CRITERIA

- DMEM[128] = 6, DMEM[129] = 21, DMEM[130] = 25
- DMEM[131] = 120, DMEM[132] = 1, DMEM[133] = 36

09 tb_power**PASS**

Source	tb/tb_power.sv	Timeout	6000 cycles
Cycles	2250	Halt	PC stable at WB stage after all eight exponentiation cases complete.

Binary exponentiation evaluates eight base/exponent pairs.

INSTRUCTION COVERAGE

MOVI LDR MUL LSR TST BEQ BNE STR INC

PASS CRITERIA

- DMEM[128] = 1048576, DMEM[129] = 2187, DMEM[130] = 15625, DMEM[131] = 16807
- DMEM[132] = 1296, DMEM[133] = 27, DMEM[134] = 4, DMEM[135] = 1

10 tb_isqrt**PASS**

Source	tb/tb_isqrt.sv	Timeout	18000 cycles
Cycles	7500	Halt	PC stable at WB stage after all ten square-root cases complete.

Integer square root calculates floor(sqrt(n)) for ten values using binary search.

INSTRUCTION COVERAGE

MOVI LDR ADD SUB LSR MUL CMP BGT BLT BEQ STR

PASS CRITERIA

- DMEM[128] = 0, DMEM[129] = 1, DMEM[130] = 1, DMEM[131] = 3, DMEM[132] = 4
- DMEM[133] = 10, DMEM[134] = 15, DMEM[135] = 100, DMEM[136] = 351, DMEM[137] = 1000

11 tb_collatz**PASS**

Source	tb/tb_collatz.sv	Timeout	20000 cycles
Cycles	9000	Halt	PC stable at WB stage after all eight Collatz cases complete.

Collatz regression counts steps to convergence for eight input values.

INSTRUCTION COVERAGE

MOVI LDR TST BEQ LSR MUL ADD INC STR BNE

PASS CRITERIA

- DMEM[128] = 0, DMEM[129] = 1, DMEM[130] = 7, DMEM[131] = 8
- DMEM[132] = 16, DMEM[133] = 14, DMEM[134] = 111, DMEM[135] = 25

12 tb_fibonacci**PASS**

Source	tb/tb_fibonacci.sv	Timeout	5000 cycles
Cycles	2400	Halt	PC stable at WB stage after all eight Fibonacci cases complete.

Iterative subroutine computes eight Fibonacci values.

INSTRUCTION COVERAGE

MOVI LDR ADD R-type BL BX R14 PUSH POP CMP BEQ BNE STR

PASS CRITERIA

- DMEM[128] = 0, DMEM[129] = 1, DMEM[130] = 1, DMEM[131] = 5
- DMEM[132] = 21, DMEM[133] = 55, DMEM[134] = 377, DMEM[135] = 4181

13 tb_bsearch**PASS**

Source	tb/tb_bsearch.sv	Timeout	30000 cycles
Cycles	14000	Halt	PC stable at WB stage after all 16 search queries complete.

Binary search over 150 sorted elements handles 16 queries: 12 hits and 4 misses.

INSTRUCTION COVERAGE

MOVI LDR indexed ADD SUB LSR mid MUL byte addr CMP BEQ BGT BLT STR BL BX R14

PASS CRITERIA

- DMEM[232..243] contains found indices {0,14,29,30,50,79,80,100,119,120,135,149}
- DMEM[244..247] contains 0xFFFFFFFF x4 for missing queries
- Total 16 queries: 12 hits + 4 misses

5 Regression Summary

The regression suite preserves the single-cycle coverage while validating the CPI=5 execution model and WB-stage architectural commit behavior.

Testbench	Description	5-Stage Cycles	Status
tb_cpu	ISA integration	425	PASS
tb_memcpy	10-word memory copy loop	330	PASS
tb_array_sum	$C[i] = A[i] + B[i]$, $N=25$	1150	PASS
tb_minmax	Min/max linear scan, $N=25$	950	PASS
tb_sort	Bubble sort ascending, $N=25$	9000	PASS
tb_factorial	$0!$ through $12!$ via BL/BX	3750	PASS
tb_bitops	AND/OR/XOR/NOT/LSL/LSR x 8 inputs	800	PASS
tb_gcd	Euclidean GCD x 6 pairs	3000	PASS
tb_power	base^{exp} binary exponentiation x 8	2250	PASS
tb_isqrt	$\text{floor}(\text{sqrt}(n)) \times 10$	7500	PASS
tb_collatz	Collatz steps-to-1 x 8	9000	PASS
tb_fibonacci	$F(n)$ iterative x 8	2400	PASS
tb_bsearch	Binary search 16 queries / 150 elements	14000	PASS

13 / 13 testbenches PASS

All testbenches PASS under Vivado 2025.1 xsim behavioral simulation. Numeric 5-stage cycle counts are shown for CPI=5.